Chalicheemala Surya Thejas

Data Science & AI Engineer | Python · Machine Learning · Full-Stack Development

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Professional Summary

Data Science and AI Engineer pursuing an Integrated MTech in Software Engineering at VIT Vellore (GPA: 9.13). Skilled in Python, machine learning, deep learning, and full-stack development using React, TypeScript, and Node.js. A collaborative team player and analytical problem-solver with strong communication skills, adept at cross-functional collaboration and delivering results under pressure. Proven track record of building and deploying AI-powered applications, computer vision systems, and end-to-end machine learning pipelines through a detail-oriented and adaptable approach.

Technical Skills

Languages: Python, Java, SQL, HTML, CSS, JavaScript, TypeScript

Frameworks & Libraries: React, Node.js, TensorFlow, Keras, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn

Cloud & Tools: Google Cloud Platform, BigQuery, Looker, Git, GitHub, Figma, VS Code

Areas of Expertise: Machine Learning, Deep Learning, Computer Vision, Data Analysis, Model Evaluation, Web Development, Artificial Intelligence, RESTful APIs

Soft Skills: Problem-Solving, Analytical Thinking, Team Collaboration, Communication, Adaptability, Attention to Detail, Time Management, Critical Thinking

Experience

NIT Calicut — **May 2026 – July 2026**

Web Development Intern — *Hybrid*

- Engineered a full-stack Kozhikode Traffic Management web application using React, TypeScript, and Node.js, enabling real-time monitoring and data visualization of city traffic flow across 10+ key corridors.
- Architected an interactive analytics dashboard integrated with Google Cloud services and RESTful APIs, cutting data ingestion latency by 35% and accelerating reporting turnaround for 5+ city planning teams.

Internship Studio — **October 2025 – March 2026**

Data Science Intern — *Remote*

- Constructed a supervised machine learning classification model using Scikit-learn, executing the complete ML pipeline – data preprocessing, feature engineering, cross-validation, and hyperparameter tuning – achieving 88%+ model accuracy.
- Assessed model performance through precision, recall, F1-score, and confusion matrix analysis, improving generalizability on unseen real-world data by optimizing decision thresholds.

Projects

- **NeuroScan MRI – Alzheimer's Risk Screening** *(Python, PyTorch, OpenCV, Gradio):* Fine-tuned ResNet-50 with attention mechanisms on 11.5K MRI scans, achieving ∼87% classification accuracy across 4 Alzheimer's stages via transfer learning and augmentation; deployed a Gradio interface reducing clinical screening effort by 60%.
- **Virtual Fitness Trainer** *(Python, OpenCV, MediaPipe, Machine Learning):* Engineered a real-time AI fitness trainer using MediaPipe pose estimation to track 33 body keypoints, auto-count exercise repetitions at 95%+ accuracy, and deliver live joint-angle feedback – eliminating the need for a human trainer.
- **Full-Stack Railway Management System** *(React, Node.js, MongoDB, REST APIs):* Architected a full-stack railway web platform serving 100+ users, with a JWT-authenticated backend achieving 40% faster data retrieval through query optimization and a responsive React frontend supporting dynamic route filtering and bookings.
- **Image CNN Classifier** *(Python, TensorFlow, Keras, Deep Learning):* Implemented a convolutional neural network attaining 92% test accuracy on 10,000 images; applied rotation, flipping, and zoom augmentation to curb overfitting and resolved class imbalance through targeted classification report analysis.

Education

Vellore Institute of Technology (VIT) — **2023 – 2028 (Expected)**

Integrated MTech – Software Engineering | GPA: 9.13 / 10.0 — *Vellore, Tamil Nadu*

Narayana Junior College — **2021 – 2023**

Board of Intermediate Education (BIE) | GPA: 9.73 / 10.0 — *Anantapur, Andhra Pradesh*

Awards & Achievements

- **VIT Hackathons (2024) – Spectra & Web-A-Thon:** Delivered 2 end-to-end AI solutions under tight deadlines – an AI Smart Watch fitness monitoring system using real-time pose estimation and ML inference (Spectra), and Skill-Sim, an NLP-driven professional conversation simulator that improved user response quality scores by 30% (Web-A-Thon).
- **Google Cloud Data Analytics (2025):** Applied BigQuery for large-scale SQL-based querying and Looker for constructing interactive BI dashboards, generating actionable insights from multi-million-row datasets on Google Cloud Platform.
- **Udemy Certifications (2024):** Certified in Full Stack Web Development (React, Node.js, Express, MongoDB) and Python for Data Science & Machine Learning (NumPy, Pandas, Scikit-learn, end-to-end ML model development).